



Options paper on the methodology for long-term temporary biogenic carbon storage in buildings.

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1 DEFINITIONS

For the purposes of this Annex, the following definitions shall apply:

- (1) 'biogenic carbon' means, according to EN 16485, carbon derived from/contained in biomass;
- (2) 'biomass' means, according to EN 16485, material of biological origin excluding material embedded in geological formations and material transformed to fossilised material;
- (3) 'bio-based construction product' means a construction product containing biomass and associated biogenic carbon,
- (4) 'building' means a roofed construction having walls and a conditioned indoor environment, and which can be categorised based on its intended purpose and form;
- (5) 'declaration of performance and conformity', or DoPC, when used in the context of this methodology, refers to the environmental essential characteristics of a construction product's life cycle, as per Regulation (EU) 2014/3110;
- (6) 'environmental product declaration', or EPD, means, according to EN 15978, an independently verified claim which indicates the environmental impacts and aspects of a product, providing quantified environmental data using predetermined parameters and, where relevant, additional environmental information;
- (7) 'estimated service life' means, according to EN 15978, the service life that a building or parts of a building would be expected to have in a set of specific in-use conditions, determined from reference service life data after taking into account any differences from the reference in use conditions;
- (8) 'new construction activity' means a construction works which has the objective to construct a new building;
- (9) 'reference building' means a hypothetical equivalent building used for an activity-specific baseline, tailored to the relevant building typology, geographical context and architectural context;
- (10) 'reference service life' means, according to EN 15978, the service life of a construction product, component, assembly or system which is known to be expected under a set of reference in-use conditions and which may form the basis of estimating the service life under other in-use conditions;
- (11) 'renovation activity' means the modification, refurbishment or retrofitting of an existing building in order to meet owner needs, occupant needs, or regulatory requirements.

2 DESCRIPTION OF THE CARBON REMOVAL ACTIVITY

2.1 Eligibility

2.1.1 Building project

Temporary carbon storage in building activities shall consist of using bio-based construction products in construction activities of buildings provided that the following eligibility criteria are met:

(a)

Option 1: The activity consists of the new construction of an office building or a building of at least four storeys, for which a standardised baseline is used;	
Option 2: The activity consists of:	
(i)	new construction of an office building or a building of at least four storeys, for which a standardised baseline is used, or;
(ii)	the new construction or renovation of any building type, provided that a corresponding activity specific baseline can be established.

- (b) bio-based construction products or materials are used in eligible building elements referred to in section 2.1.2;
- (c) the building and bio-based construction products are compliant with the sustainability criteria in section 6;
- (d) the expected service life of the building is at least 35 years.

To avoid double counting, any biochar used in construction products that is associated with the issuance of permanent carbon removal units shall not be considered as eligible for counting as temporary carbon storage.

2.1.2 Building elements

Eligible building elements shall have an estimated service life of at least 35 years and not be at risk of removal during normal use and minor renovations of the building. By default, eligible building elements shall be those listed below, based on tier 3 elements listed in Annex III to Directive (EU) 2024/1275:

- Foundations (substructure, up to and including ground floor slab);
- Basement elements;
- Structural frames, slabs and panels (situated above ground floor slab and including roof structure elements);
- Basement insulation;
- Roof insulation;
- Floor, ceiling and wall insulation.

2.2 Activity period, monitoring period and certification period

2.2.1 Activity period

The activity period corresponds to the building's construction works, during which biogenic carbon is effectively stored through the incorporation of bio-based products into eligible building elements. The finalisation of the activity period shall match with the date of completion of the renovation or construction works.

2.2.2 Monitoring period

The monitoring period shall start at the same time as the activity period, and its duration shall be at least 35 years longer than the activity period.

Option 1: Operators may commit to a monitoring period at the time of submission of the activity plan of either 35, 50 or 100 years.

Option 2: Operators may commit to a monitoring period at the time of submission of the activity plan of either 50 or 100 years.

2.2.3 Certification period

The certification period shall match the activity period, and a recertification audit shall be conducted upon completion of the activity period.

2.3 Planning and reporting

2.3.1 Activity plan

Before the certification audit, the operator shall submit to the certification body an activity plan that includes the information necessary to assess compliance with the requirements of this methodology.

The activity plan shall include:

1. A general description of the activity (construction and/or renovation), which includes:
 - (a) Details of the building or building portfolio it applies to (for example typology, useful interior floor area(s), estimated service life(lives)).
 - (b) Legal ownership and contact information of the operator;
 - (c) Address(es) and cadastre reference(s) of the building(s) associated with the activity;
 - (d) A general description of the construction or renovation project;
 - (e) The start date of the activity;

- (f) Any information relating to participation of the operator or their legal predecessor with other certification schemes during the last 5 years for the same type of activity.
- 2. A description of how the certification methodology will be applied in the activity:
 - (a) Identification of the relevant baseline (standardised or activity-specific) based on the building typology or reference building.
 - (b) Information management process for tracking quantities of eligible materials throughout the activity. Definition of roles and responsibilities of product manufacturer, supplier (if not the manufacturer), contractor record-keeping for site deliveries, onsite waste and any over-ordered materials, and estimations of materials installed in the building(s) by the architect.
 - (c) How the DoPC or EPD shall be used for calculations of biogenic carbon storage and GHG-associated, if any supplementary data or assumptions to cover life cycle modules A4 and A5 will be necessary, and if so, how this information will be gathered;
 - (d) In the case of activity-specific baselines, how the choice of eligible bio-based construction products is more expensive than the reference building scenario;
 - (e) How compliance with relevant sustainability requirements will be achieved.
- 3. Estimates of total temporary carbon removals and GHG-associated, based on product-level DoPC or EPD data and multiplied by the expected quantities of eligible products to be used in the activity.
- 4. Estimates of the net carbon removal benefits generated by the activity.
- 5. Any other information necessary for the certification body to conduct the certification audit in accordance with Article 9 of Regulation (EU) 2024/3012.

2.3.2 Monitoring plan

Before the certification audit, operators shall submit a monitoring plan to the certification body. That monitoring plan shall include:

- (a) a description of the building to be monitored, including technical drawings of the building site layout, floor plans, cross-sections, wall sections, exterior elevations, interior elevations and roof plans;
- (b) a bill of quantities for eligible bio-based construction products used in eligible building elements, with quantities and volumes clearly indicated in building drawings;
- (c) the derivation, if applicable, of any project-specific GWP factors for reporting on GHG-associated with the transport of eligible bio-based construction products to the building site (module A4) or of onsite waste generation from eligible bio-based construction products during installation (module A5);
- (d) a calculation for the balancing of purchased quantities of eligible bio-based construction products against the quantities actually incorporated into eligible building elements – the remainder must either be: (i) used in non-eligible building elements, (ii) discarded as waste, or (iii) remaining as over-ordered material. In case of the latter point, the fate of the over-ordered eligible material shall be documented;
- (e) a description of the procedure for monitoring and reporting in cases of any changes to the building that could potentially affect the quantities of eligible bio-based construction products remaining in eligible building elements during its use phase.

In the case that it is not possible to fully detail the monitoring plan when an operator applies for certification, the monitoring plan shall be submitted as completely as possible, clearly indicating any non-final aspects and providing an indication of how the operator expects these aspects to be addressed. The activity may be certified on this basis provided the certification body accepts that the omissions are properly justified. The monitoring plan shall be finalised and presented to the certification body prior to the first and only re-certification audit.

Certification schemes may provide additional guidance on the elements to be included in the monitoring plan.

2.3.3 Monitoring report

Operators shall obtain, record, compile, analyse and document monitoring data, including assumptions, references, activity data and calculation factors in a transparent manner that enables the checking of performance achieved during the certification period, when requested, and report this information to the certification bodies or certification schemes.

Certification schemes shall allow for the sharing of monitoring data in an anonymised way with the Commission upon request by competent national authorities or the Commission

2.3.3.1 Monitoring report before the re-certification audit

For the purpose of the re-certification audit, the monitoring report shall include:

- (a) a description of the as-built or finally renovated building, including final signed versions of all technical drawings indicated in the monitoring plan;
- (b) a final bill of quantities for eligible bio-based construction products used in eligible building elements, signed by the project architect and with quantities and volumes clearly indicated in building drawings;
- (c) if applicable, an explanation for any project-specific GWP factors for reporting on GHG associated with the transport of eligible bio-based construction products to the building site (module A4) or of onsite waste generation from eligible bio-based construction products during installation (module A5);
- (d) presentation of the stored biogenic carbon values and GHG associated data from relevant DoPCs or EPDs, and their multiplication, from declared functional units, up to bill of quantities incorporated into eligible building elements;
- (e) a calculation for the balancing of purchased quantities of eligible bio-based construction products against the quantities actually incorporated into eligible building elements – the remainder must either be: (i) used in non-eligible building elements, (ii) discarded as waste, or (iii) remaining as over-ordered material. In case of the latter point, the fate of the over-ordered eligible material shall be documented;
- (f) if an activity-specific baseline is used, the description of the reference building used and information to verify compliance with the additionality tests laid down in section 4;
- (g) the results of the risk assessment screening laid down in section 5.1;
- (h) information to verify compliance with the relevant sustainability requirements laid down in Section 6.

2.3.3.2 Monitoring report before monitoring audits

For the purposes of the monitoring audits, the operator shall submit the monitoring report at least every five years and within one year of becoming aware of an event resulting in a reversal or any change in operator.

In the absence of any reversals or any change in operator, the monitoring report shall consist of a statement which confirms the continued existence of the building and the ongoing presence of the certified carbon storage in the building.

If the operator has changed during the monitoring period, the original operator shall retain liability unless it is agreed with the new operator for liability for certified units to be transferred to them. In such cases, the new operator shall be registered under the certification scheme and assume the role of operator and associated responsibilities, including the production of future monitoring reports.

If a reversal has occurred during the monitoring period, the monitoring report shall:

- (a) include a description and relevant data related to the observed reversal event and classify the event as being avoidable or unavoidable (see section 5.1);
- (b) in cases where the reversal is partial, quantify the amount of the reversal by describing in sufficient detail the scope of the reversal;
- (c) in the cases of partial reversal events, a description of measures undertaken to limit the extent of the reversal and prevent or minimise similar events occurring in the future.

In all cases, certification bodies may request additional information, such as photographic evidence or public registry records, to support claims made in the monitoring report.

3 QUANTIFICATION OF BASELINE, TOTAL CARBON REMOVAL AND ASSOCIATED GHG EMISSION

3.1 Data sources and requirements

Calculations and reporting shall be based on product-level data that is multiplied up to building level based on the quantities of products used in eligible building elements.

3.1.1 Product level data

Data relating to GHG sources and sinks shall be product-specific and, if available in more than one format, shall be preferred in the following order:

- Product-specific data of climate change effects obtained from a DoPC under Regulation (EU) 2024/3110, including DoPCs covered by a harmonised technical specification and DoPCs issued in accordance with a relevant European assessment document and European technical assessment.
- Product-specific data from an EPD calculated in accordance with standard EN 15804 or a compatible standard, although not issued under Regulation (EU) 2024/3110.

3.1.1.1 Temporal validity of data

All eligible products for which temporary carbon storage is claimed must be covered by DoPCs or EPDs that are valid at the date of purchase of the materials during the activity period.

If the same materials are purchased in batches over a prolonged period during which the DoPC or EPD is updated, the updated data shall be applied to the newly purchased products, and the earlier data shall be maintained for the previously purchased products.

3.1.2 Building level data

Calculations at building level shall respect the eligibility conditions set out in section 2.1.

Stored carbon values used for building-level CRCF calculations shall be exclusively based on carbon stored in eligible construction products that are used in eligible building elements. Calculations shall not consider:

- (a) any biogenic carbon stored in construction product packaging
- (b) any biogenic carbon stored in eligible bio-based construction products used in non-eligible building elements;
- (c) any atmospheric carbon uptake by vegetation located in, on or around the building(s) subject to the activity;
- (d) any projected atmospheric carbon uptake during the use phase or end-of-life phase of the building(s) subject to the activity via the carbonation of alkaline construction products;

3.2 Quantification of the net removals of the activity

Quantification of the net removals shall be based on declared information at the level of CRCF-eligible construction products or elements in terms of the carbon removals and GHG emission per functional unit, and at the level of construction/renovation project in terms the numbers of functional units per project and per m2 floor area of the building.

Construction product information shall preferentially be based on a declaration of performance and conformity (DoPC) in line with harmonised reporting standards under the Regulation (EU) 2024/3110. When a DoPC is not available, this information shall alternatively be provided from valid, third-party certified Environmental Product Declarations (EPDs) that are in line with EN 15804 or a compatible standard.

3.2.1 Overall quantification equation

The quantification equation set out in Article 4(5) of Regulation (EU) 2024/3012 shall apply as detailed below:

$$TNCRB = CR_{baseline} - CR_{total} - GHG_{associated}$$

Where:

- $TNCRB$ = temporary net carbon removal benefit, in units of tonnes CO₂eq.
- CR_{total} = the quantity of biogenic carbon stored in eligible bio-based construction products in eligible building elements, in units of tonnes CO₂eq. as detailed in section 3.2.1.1
- $GHG_{associated}$ = the sum of GWP (modules A1-A5), associated with the eligible bio-based construction products used in eligible building elements, in units of tonnes CO₂eq., as detailed in section 3.2.1.2
- $CR_{baseline}$ = the quantity of biogenic carbon stored in bio-based construction products in eligible building elements, in units of tonnes CO₂eq. and adjusted for baseline $GHG_{associated}$, as detailed in section 3.2.1.3

3.2.1.1 Biogenic carbon storage

The biogenic carbon storage, in units of kgCO₂/functional unit, shall be reported in the DoPC or EPD based on:

- The mass of biomass per functional unit.
- The density of the biomass material.
- The average moisture content of the biomass.
- The average dry mass-based elemental carbon content of the biomass.

An example calculation for calculating the biogenic carbon stored in a timber EPD for a functional unit of 1m³, having a density of 450 kg/m³, moisture content of 12% and dry mass elemental carbon content of 50% would be:

$$Carbon\ storage\ (kgCO_2) = 1(m^3) \times 450(kg/m^3) \times \frac{1}{(1 + 0.12)} \times 0.50(kgC/kgDM) \times \frac{44}{12}$$

Where:

- 0.12 is the decimal of the 12% moisture content,
- 0.50 is the decimal of the 50% share of elemental carbon (C) content in dry matter (DM), and
- 44/12 is the molar mass ratio of CO₂/C (both in units of g/mole which cancel out).

Numerically, the calculation would first convert 1m³ into 450kg of mass, then adjust this to dry mass (approximately 402kg), and then to mass of elemental carbon (approximately 201kg) before converting it into CO₂ (approximately 737kg CO₂). The biogenic carbon storage would effectively be 737 kgCO₂/m³ timber.

Biogenic carbon storage for the eligible bio-based construction product, and for packaging used for the same product, shall be separately calculated and reported in DoPCs or EPDs.

3.2.1.2 GHG-associated emissions

Option 1: Data relating to the processes, sub-processes and reference flows per functional unit of eligible product shall be expressed in terms of GWP_{total} , GWP_{fossil} , $GWP_{biogenic}$ and GWP_{LULUC} in line with harmonised technical specifications for DoPCs, with EN 15804-compatible standards, or with EN 15804, and for the modules in the table below.

The GHG-associated shall be counted as the sum of the following GWP emissions:

$$GHG_{associated} = GWP_{fossil(A1-A5)} + GWP_{biogenic(A2-A5)} + GWP_{LULUC(A1-A5)}$$

Option 2: Data relating to the processes, sub-processes and reference flows per functional unit of eligible product shall be expressed in terms of GWP_{fossil} in line with harmonised technical specifications for DoPCs, with EN 15804-compatible standards, or with EN 15804, and for the modules in the table below.

The GHG-associated shall be counted as the sum of the following GWP emissions:

$$GHG_{associated} = GWP_{fossil(A1-A5)}$$

Table 1. Life cycle modules for GHG-associated calculations and related remarks.

Life cycle module	General remarks	GHGassociated remarks ⁽¹⁾
A1: Extraction and upstream production	Includes the impacts of processes associated with cultivating and harvesting the raw materials.	This module is where the uptake of atmospheric carbon and its conversion into biogenic carbon takes place.
A2: Transport to factory	From raw material production or, from the point of waste collection in the case of recycled wastes, to the factory related to A3.	GHGassociated will be dominated by the use of fossil fuels combusted in vehicle engines or electricity consumption via electrified transport modes.
A3: Manufacturing	Includes all processes associated with the operation of the production line for the eligible bio-based construction products.	GHGassociated will include impacts from consumption of electricity, chemicals, water, emissions to air, wastewater and solid waste.
A4: Transport	From the manufacturing factory to the customer site.	In a DoPC or EPD, the producer will have many potential customers and so the impacts here will represent an average transport scenario. In cases where this information is not provided, or is deemed inaccurate for the activity, project-specific estimates shall be used for the activity (see section 3.1.2).
A5: Construction installation process	Impacts associated with the installation of the eligible bio-based construction product into eligible building elements.	This module potentially includes impacts associated with ancillary materials used for installation (e.g. adhesives, brackets etc.). Impacts associated with packaging waste and waste from offcuts are counted here. From the producer perspective, this would be an average scenario. In cases where this information is not provided, or is deemed inaccurate for the activity, project-specific estimates shall be used (see section 3.1.2).
(1) The remarks included here are general and, if they in any way contradict rules in relevant harmonised technical specifications for DoPCs, or relevant EN standards for EPDs, then these shall take precedence.		

Data reported in DoPCs or EPDs for modules A1, A2 and A3 may be bundled together if a breakdown of this information is considered as potentially revealing confidential information about suppliers and supply chains.

All impacts associated with modules A1 to A5 shall be normalised per functional unit (e.g. m³, tonne etc.) of the eligible bio-based construction products. These functional units shall be used in calculations at building level described in section 3.1.2.

3.2.1.3 Baselines

Option 1:

A standardised baseline equal to zero shall apply to the new construction of office buildings or buildings of at least four storeys.

Option 2:

New office buildings and buildings of at least four storeys

A standardised baseline equal to zero shall apply to the new construction of office buildings or buildings of at least four storeys.

Other buildings

For all renovation activities and new construction activities of other building types, activity specific baselines shall be used.

The reference building used for an activity-specific baseline shall be based on the same building design used in the activity, but with a “business as usual” choice of construction products in the eligible building elements based on similar buildings in the same geographical context. The eligible building elements in the reference building shall be:

- Based on the same or similar building envelop geometry.
- Based on the same or similar useful floor area and internal partitions for the whole building.
- Based on a similar loadbearing capacity for reference structural elements.
- Based on a similar thermal performance for reference insulation materials.

The stored biogenic carbon and GHG associated with the eligible building elements shall be compared to the stored biogenic carbon and GHG associated in the same eligible building elements in the building subject to the activity. The activity shall generate a temporary net carbon removal benefit when:

$$\left(\sum CR_{baseline} - \sum GHG_{baseline} \right) < \left(\sum CR_{total} - \sum GHG_{total} \right)$$

Where:

- $CR_{baseline}$ is the stored biogenic carbon in any eligible bio-based construction products, expressed as kgCO₂eq. and with a positive sign, that would be used in eligible building elements in the reference building.
- $GHG_{baseline}$ are the GHG associated, as detailed in section 3.2.1.2, expressed as kgCO₂eq. and with a positive sign, that would be used in eligible building elements in the reference building.
- CR_{total} is the stored biogenic carbon in construction products, expressed as kgCO₂eq. and with a positive sign, that are used in eligible building elements in the activity.
- GHG_{total} are the GHG associated, as detailed in section 3.2.1.2, expressed as kgCO₂eq. and with a positive sign, that are used in eligible building elements in the activity.

Values for GHG associated shall be obtained from generic EPDs for the reference construction products that would be used in the hypothetical reference building and may be directly specified by the certification scheme. In cases where no eligible bio-based construction products are used in eligible building elements in the reference building, the activity-specific baseline shall be set to zero.

3.3 Uncertainties and data quality

The quality level of product-level DoPCs-~~or~~ EPDs shall be assessed against the criteria defined in the table below for geographical, technical, and time representativeness.

Table 2. Data quality level and criteria from the Product Environmental Footprint Category Rules

Quality level	Geographical representativeness	Technical representativeness	Time representativeness
Very good	The processes included in the data set are fully representative for the geography stated in the “location” indicated in the metadata	Technology aspects have been modelled exactly as described in the title and metadata, without any significant need for improvement	Data are not older than 0 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)
Good	The processes included in the data set are well representative for the geography stated in the “location” indicated in the metadata	Technology aspects are very similar to what described in the title and metadata with need for limited improvements. For example: use of generic technologies’ data instead of modelling all the single plants.	Data are not older than 3 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)
Fair	The processes included in the data set are sufficiently representative for the geography stated in the “location” indicated in the metadata. E.g. the represented country differs but has a very similar electricity grid mix profile	Technology aspects are similar to what described in the title and metadata but merits improvements. Some of the relevant processes are not modelled with specific data but using proxies.	Data are not older than 6 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)
Poor	The processes included in the data set are only partly representative for the geography stated in the “location” indicated in the metadata. E.g. the represented country differs and has a substantially different electricity grid mix profile	Technology aspects are different from what described in the title and metadata. Requires major improvements.	Data are not older than 10 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years, confirmed by the reviewer(s))
Very poor	The processes included in the data set are not representative for the geography stated in the “location” indicated in the metadata.	Technology aspects are completely different from what described in the title and metadata. Substantial improvement is necessary.	Data are older than 10 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)

Option 1: The data quality level shall be at least “fair” to be used for the quantification under this methodology.

Option 2:

The data quality shall be at least fair and better quality data shall be incentivised by the use of a layered approach to further distinguish between “fair”, “good” and “very good” quality data. Data that overall tends towards a “very

good” quality level would be accepted without further changes, but as data quality tends towards “fair”, an uncertainty factor would proportionately apply to increase the assumed GHG associated, up to a maximum of 3%.

Table 3. Data quality level and criteria from the Product Environmental Footprint Category Rules

Quality level	Geographical representativeness	Technical representativeness	Time representativeness
Very good	No penalty	No penalty	No penalty
Good	-0.5%	-0.5%	-0.5%
Fair	-1%	-1%	-1%
Poor	Not permitted	Not permitted	Not permitted
Very poor	Not permitted	Not permitted	Not permitted

4 ADDITIONALITY

Option 1: As only standardised baselines are used under this methodology, additionality as referred to the CRCF Regulation shall be considered to be complied with.

Option 2: In cases where an activity-specific baseline is applied, the rules related to a reference building scenario, regulatory test and financial viability shall be applied.

4.1 Regulatory test

This test shall demonstrate that the net climate benefit resulting from the activity does not occur as a result of any legal requirement on the operator stemming from Union or national legislation to carry out the activity in such a way as to use a certain quantity of eligible bio-based construction products in eligible building elements, unless such legislation refers to or formally integrates certification under Regulation (EU) 2024/3012 as an instrument for implementation.

National legislation includes laws, statutes and regulations also at regional level, as well as court orders, mandatory environmental management agreements or planning decisions or other legally binding agreements.

Activities going beyond the level required under national or Union obligations may generate units eligible for certification for those additional emission reductions or carbon removals.

4.2 Financial viability

The incentive effect of the certification of the activity shall be analysed using a simple cost test for a comparison of the CAPEX of the construction or renovation of eligible building elements against the hypothetical CAPEX of the same elements in the reference building.

The simple cost test shall be conducted using the following steps:

- Describing the CAPEX costs of the eligible building elements ⁽¹⁾ in the activity.
- Describing the CAPEX costs of the same building elements in the hypothetical reference building.

The scope of CAPEX accounting shall at least include material costs. Certification schemes may also require other cost elements to be included in a consistent manner, such as transport costs, equipment hire costs and labour costs. Certification schemes may also provide default CAPEX values for reference buildings that are normalised on a per m2 basis and that can be used as an initial gauge for the financial viability test.

When $CAPEX_{activity} > CAPEX_{reference}$, the financial incentive effect shall be considered as valid.

¹ As described in section 2.1.2, eligible building elements are: - Foundations (substructure, up to and including ground floor slab); Basement elements; Frames, slabs and panels (structure, above ground floor slab and including roof structure elements); Basement insulation; Roof insulation; Floor, ceiling and wall insulation.

5 STORAGE, MONITORING AND LIABILITY

5.1 Risk assessment

Operators shall carry out a risk assessment for the activity before the start of the activity period. The risk rate resulting from the risk assessment of the activity shall account only for unavoidable events.

Reversals resulting from the management of the activity and any other wilful actions by the operator or from mismanagement, neglect or illegal actions by the operator, including as a consequence of insolvency, or reversals that have occurred following a failure to implement the risk mitigation measures that is attributable to the operator, shall be classified as avoidable.

Reversals resulting from natural events and extreme events, war or acts of terrorism, changes in policies or legal requirements that prevent the operators from implementing risk mitigation measures and reversals resulting from illegal action by third parties that may not be controlled, influenced or managed using legal means by the operator shall be classified as unavoidable.

Reversals caused by any factors that were not identified in the risk assessment shall be classified as avoidable and may only be classified as unavoidable with due justification, such as in cases of clear *force majeure*.

Operators shall update the risk assessment after any unavoidable event.

5.1.1 Climate-related risks at the building location

Operators shall, based on the location of the building and its features, conduct a screening exercise for the climate-related risks that could result in the reversal of carbon storage during the monitoring period listed below.

Climate-related risk to buildings	Risk level and associated additional risk rate (% of certified units)				Mitigated in order to reach "low"?
	Insignificant	Low	Medium	High	
Fluvial flooding	n/a	0,5 %	Project rendered ineligible for CRCF certified units		Yes / No
Storm surges		0,5 %			Yes / No
Coastal flooding		0,5 %			Yes / No
Coastal erosion		1,0 %			Yes / No
Avalanches		1,0 %			Yes / No
Landslides		1,0 %			Yes / No
Subsidence		0,5 %			Yes / No
Glacial lake, reservoir or lagoon outbursts		1,0 %			Yes / No

Results for each risk shall be categorised as either: insignificant, low, medium or high. A "low" risk result must be obtained for all climate-related risks. In cases where a "high" risk level is identified, the building shall be considered as not being situated in an appropriate location. In cases where a medium result is obtained, the implementation of risk mitigation measures that would reduce the risk level to "low" may be permitted by the certification scheme.

The certification scheme may provide additional guidance on the approach and categorisation of climate-related risks and associated mitigation measures.

5.1.2 Building-level risks

Operators shall conduct a qualitative screening risk assessment of potential moisture damage to bio-based construction products used in eligible building elements that is based in the intended building use and specific features of the building.

Risk type	Description of building or site works feature	Risk rate if applicable and unmitigated (% units)	Applicable?	Mitigated?
Moisture ingress (via roof)	Use of low roof slope(s)	0,2 %	Yes / No	Yes / No
	Check on weatherproofing of roof	0,2 %	Yes / No	Yes / No
	Check on roof drainage infrastructure	0,2 %	Yes / No	Yes / No
	Details of any green roof	1,0 %	Yes / No	Yes / No
Moisture ingress (via floor or walls)	Accessibility of hot- and cold-water pipe connections and tanks	0,5 %	Yes / No	Yes / No
	Details of any indoor swimming pool or other high-humidity areas	1,0 %	Yes / No	Yes / No
	Check on cavity ventilation	0,5 %	Yes / No	Yes / No
	Details of any green wall	1,0 %	Yes / No	Yes / No
Moisture ingress during works	Appropriate storage and protection of construction products onsite (prior to and during installation)	1,0 %	Yes / No	Yes / No
	Allowance for sufficient ventilation of indoor spaces after installation.	0,5 %	Yes / No	Yes / No

5.1.3 Storage period-related risks

Certified units for the activity shall be issued upon completion of the one-off re-certification audit when the construction or renovation works are completed. Certification schemes may certify temporary storage activities as having claimed storage for a minimum period of 35 years, 50 years or 100 years.

Certification schemes shall apply the risk rates below as a function of the claimed storage period:

- 35-year storage claim: 3% buffer commitment of certified units;
- 50-year storage claim: 6% buffer commitment of certified units;
- 100-year storage claim: 15% buffer commitment of certified units.

5.2 Liability

For activities that generate a temporary net carbon removal benefit, a liability mechanism shall be used.

To that end, operators shall either participate in a buffer pool managed by a certification scheme or obtain and maintain sufficient coverage under an insurance policy or comparable guarantee products. The insurance company or comparable guarantee shall fulfil the relevant financial market rules and shall manage a pool of units.

Certification schemes or insurance companies shall allocate to the buffer pool or pool of units a share of certified units that is equal to the sum of the risk rates identified in sections 5.1.1, 5.1.2 and 5.1.3. In the event of a reversal event, the extent of the liability shall be calculated by using the tonne-year concept to determine the share of total certified units which are liable. Some worked examples are provided in Appendix III.

The certification scheme or insurance company shall ensure that, in the event of an unavoidable and avoidable reversal of the carbon underlying units that have been retired, these units are replaced by an equal number of units from the buffer pool or the pool of units. Such units shall be valid for the same period of time as the units that have been retired. Under an insurance policy or comparable guarantee product, such replacement may not take the form of monetary compensation.

For avoidable reversals for which units have been cancelled from the buffer pool or pool of units, operators shall be fully liable for replenishing the buffer pool or pool of units with an equivalent number of units of the same duration as the units that have been cancelled.

Early termination of a carbon storage in buildings activity shall be considered equal to a full reversal.

Certification schemes or insurance companies shall ensure the resilience, sufficiency, and solvency of the buffer pool or pool of units, including through regular stress-testing of the sufficiency of risk coverage. Certification schemes or insurance companies shall publish annually the composition of the buffer pool or pool of units, including the share of units allocated to the pool by vintage.

The stress test shall assess the resilience of the pool to a range of reversal risk scenarios based on, inter alia, the range of risk ratings as well as significant loss events, affecting the activities linked to the pool. The stress test shall occur at least every five years and in the event of a significant loss event. Where necessary based on the results of the stress test, certification schemes or insurance companies shall undertake measures to address risks affecting the resilience, sufficiency and solvency of the buffer pool or pool of units.

After the end of the monitoring period, the buffered units shall expire.

6 SUSTAINABILITY

6.1 Minimum sustainability requirements

During the activity period, operators shall demonstrate compliance the minimum sustainability requirements set out in sections 6.1.1 to 6.1.6.

6.1.1 Protection and restoration of biodiversity and ecosystems

Operators shall ensure that any eligible wood-based construction products are covered by a due diligence system and chain of custody certificates issued by an independent third-party certification scheme. The due diligence system shall ensure that wood-based materials are legally sourced and that a risk assessment approach is used to minimise the possibility of human right violations, damage to high conservation value forests and deforestation. At least 70% of the material in eligible wood-based construction products shall originate from virgin material from sustainably managed forests and/or recycled wood-based materials.

6.1.2 Climate Change Adaptation

No minimum requirements apply to renovation activities. For new construction activities, if the new development is on a greenfield site, the operator shall ensure that the peak rate of storm water discharge from the site does not exceed the peak rate that would occur from the same site in its pre-development condition. This shall be verified using hydrological and hydraulic modelling software against an appropriate design storm. The design storm should be for a return period of at least 1 in 50 years and use design storm hyetographs that are endorsed by the national meteorological authority or a relevant expert.

6.1.3 Transition to a circular economy

The operator shall ensure that the eligible bio-based construction products are not combined with different materials (e.g. bonded insulation foam) or treated with chemicals (e.g. wood preservatives, flame retardants etc.) during their production or installation that would compromise their recyclability in the future.

6.1.4 Pollution prevention and control

Operators shall ensure that any eligible bio-based insulation materials emit less than 0,06 mg of formaldehyde per m³ of material or component upon testing in accordance with the conditions specified in Annex XVII to Regulation (EC) No 1907/2006 and less than 0,001 mg of other categories 1A and 1B carcinogenic volatile organic compounds per m³ of material or component, upon testing in accordance with EN 16516 or ISO 16000-3:2011 or other equivalent standardised test conditions and determination methods.

6.1.5 Climate Change Mitigation

No minimum requirements apply.

6.1.6 Sustainable use and protection of water and marine resources

No minimum requirements apply.

Appendix I: Mapping of CRCF plans, reports and audits against activity timeline

The activity of carbon storage in buildings requires a tailored approach to the administrative requirements of the CRCF Regulation that fits with the nature of the activity and ensures that burdens do not become disproportionate for operators and certification bodies. The proposed approach is illustrated below.

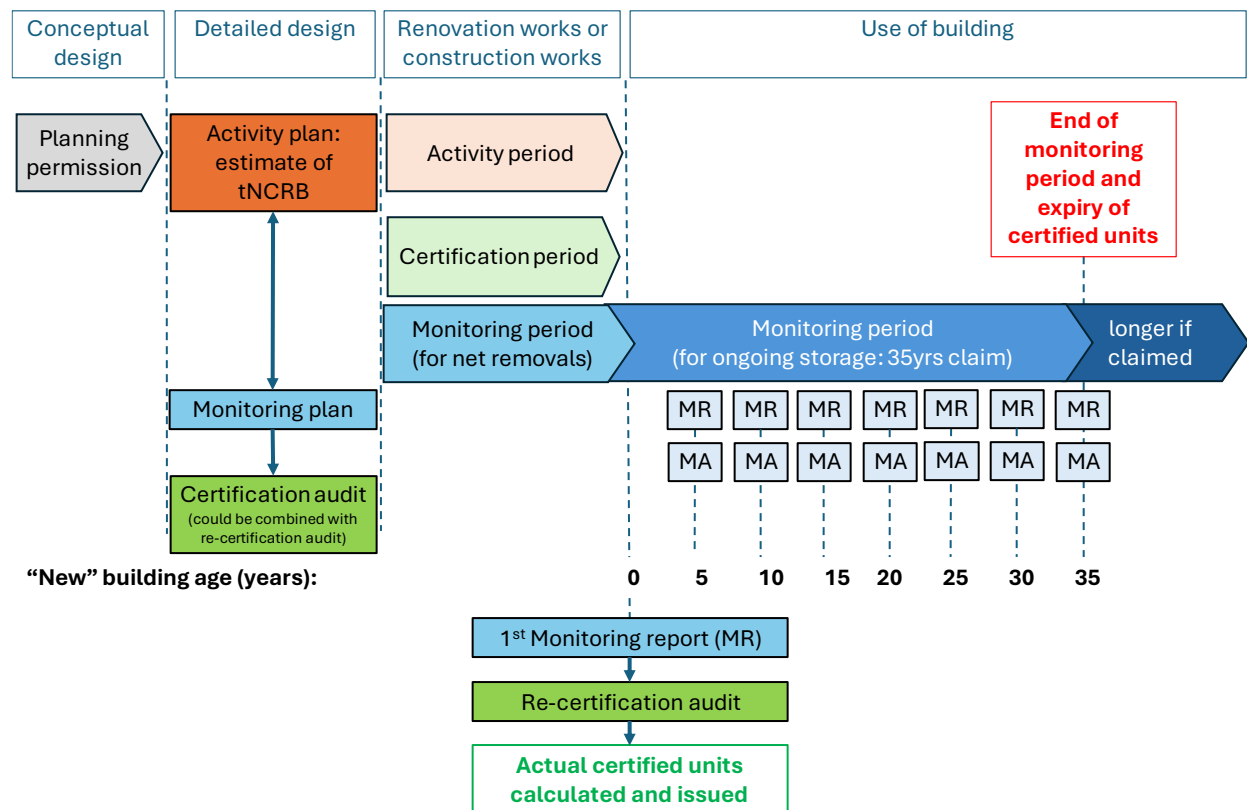


Figure 1. Visual illustration of required CRCF plans, audits and reports against the building life cycle

As shown above, the activity plan and monitoring plan should be in place before the works commence. These plans would be reviewed as part of the certification audit, but shall consist of estimates of what is going to happen during the construction or renovation works.

The construction or renovation works are synonymous with the activity period and certification period. It is during this stage that all eligible bio-based construction products would be locked into the eligible building elements. The actual eligible products used and the associated quantities will be reflected in the 1st monitoring report. This report is used as the basis for the re-certification audit and the determination of the number of certified units to be issued.

Since the activity is effectively a one-off operation, there is only a need for one re-certification audit. During the use phase of the building, monitoring reports and audits shall consist of checking against possible reversals that may have happened due to damage or deliberate modification of eligible building elements in the building structure. This should happen at a regular frequency of five years. Once the claimed storage period is reached, the certified units would then expire and be removed from the registry.